SWGI → Red Hat Validation Preparation (Step 1)

This document outlines the **technical validation preparation phase** required to move the current SWGI demo environment toward Red Hat / OpenShift readiness. The goal of this step is **not full production hardening**, but achieving a **clean validation baseline** so the container and deployment model meet Red Hat ecosystem technical expectations.

Objective

Convert the current SWGI demo container into **Red Hat-compatible container image and OpenShift-ready deployment package**, ensuring it passes validation tools such as Preflight and follow OpenShift runtime standards.

This phase focuses on:

• Container compliance • Security posture • OpenShift runtime compatibility • Validation scanning • packaging for OpenShift deployment

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Phase 1 — Container Base Image Compliance

Update the SWGI container build to use a Red Hat supported base image.

Tasks:

• Replace the current base image with:

`registry.access.redhat.com/ubi9/ubi-minimal`

• Ensure the container only installs runtime dependencies required for execution.

• Use a **multi-stage build** if necessary so development tools are not included in the final image.

• Confirm the final image is minimal and production-safe.

Deliverable:

• Updated Dockerfile using UBI minimal

Phase 2 — OpenShift Runtime Compatibility

Ensure the container follows OpenShift runtime constraints.

Tasks:

- The container must **run as non-root**.
- Remove any root-only assumptions.
- Confirm file writes occur only in writable directories such as:

`/tmp`

- Ensure no privileged container permissions are required.
- Confirm the container can run under **OpenShift restricted SCC**.

Deliverable:

- Container runs successfully with non-root user

Phase 3 — Required Health and Metrics Endpoints

Verify that operational endpoints exist and function correctly.

Endpoints expected:

- `/healthz` — container health status
- `/metrics` — Prometheus metrics endpoint
- `/authorize` — SWGI authorization gate
- `/receipt/{receipt_id}` — trust receipt retrieval
- `/receipts/export.csv` — audit export

Deliverable:

• All endpoints accessible and functional

Phase 4 — Image Metadata Labels

Add Red Hat compatible container metadata labels to the Dockerfile.

Expected labels include:

• `name`

• `vendor`

• `version`

• `release`

• `summary`

• `description`

These labels are required for ecosystem validation tooling.

Deliverable:

• Dockerfile updated with proper metadata labels

Phase 5 — Preflight Validation Scan

Run the Red Hat **Preflight** validation tool against the container image.

Tasks:

• Execute preflight scan against the built container image.

• Resolve all validation failures.

• Address:

• image metadata issues

• security warnings

• container runtime compatibility

• Repeat scan until validation passes.

Deliverable:

• Successful Preflight scan result

Phase 6 — OpenShift Deployment Packaging

Provide a minimal deployment package so the container can be installed in OpenShift.

Recommended approach for this phase:

Helm chart (fastest path).

The Helm chart should include:

• Deployment

• Service

• Ingress / Route

• Configurable environment variables

• Resource requests and limits

• Health probe configuration

Deliverable:

• Helm chart capable of deploying SWGI service to OpenShift

Phase 7 — RBAC and Service Account

Implement least-privilege service configuration.

Tasks:

• Create dedicated ServiceAccount for the SWGI service.

• Assign only required permissions.

• Avoid cluster-admin or overly broad privileges.

Deliverable:

• Minimal RBAC configuration included in deployment

Phase 8 — Configuration Management

Ensure configuration is externalized.

Tasks:

• Move configuration into:

• environment variables

• ConfigMaps

• Secrets (if applicable)

• Remove hardcoded credentials or configuration values.

Deliverable:

• Container runs with external configuration

Phase 9 — Documentation Artifacts

Provide minimal documentation required for validation review.

Required items:

• README.md

• OpenShift deployment instructions

• Architecture overview

• Environment variables documentation

• Version information

Deliverable:

• Basic documentation package for reviewers

Expected Outcome of Step 1

After completing this phase the SWGI container should be:

• Built on Red Hat UBI base image

• Compatible with OpenShift runtime restrictions

• Passing Preflight validation checks

• Deployable through Helm

• Running non-root

• Equipped with health and metrics endpoints

This establishes a **Red Hat validation-ready baseline** before proceeding to the full production packaging and ecosystem listing stages.

Next Phase (Step 2)

After validation readiness:

• Expand deployment architecture

• Add full UI demo layer

• Implement federation nodes

• Prepare marketplace packaging

• Expand documentation for enterprise deployment